# DiMuto: Innovating with blockchain, AI, and Google Cloud to change the way we track fruits

DiMuto created a
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AI Hub and block
empower farmer
Malaysia to Mexi
enable transpare
commerce.

## About DiMuto

DiMuto is a Singapore-based agritech venture that enables fair and sustainable global trade of food products through a blockchain and AI image analysis platform. Its mission is to empower smallholder farmers while overcoming food waste. Since its founding in 2018, DiMuto has built a global network across seven nations, with plans to expand into Europe, Africa, and South America.

Industries: Agriculture, Forestry & Fishing, Technology

Location: Singapore

## Google Cloud results

- Powers groundbreaking blockchain platform with AI and Google Cloud solutions
- Facilitates tracking of 100 million fruit in global network from Asia to the Americas
- Guarantees integrity of $100 million worth of worldwide agriculture shipments

Google Cloud Platform (https://cloud.google.com/)

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AI Hub (https://cloud.google.com/ai-hub/)

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BigQuery (https://cloud.google.com/bigquery/)

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AI Platform (https://cloud.google.com/ai-platform/)

TensorFlow (https://www.tensorflow.org/)

globa

TensorFlow (https://www.tensorflow.org/)

Millions of crates of fruit are shipped to global markets every day, but they face destruction if even one item inside is found to be defective.

Roughly a third of food produced for human consumption every year, some 1.3 billion tons, is lost or wasted, according to the UN Food and Agriculture Organization (FAO).

(http://www.fao.org/save-food/resources/keyfindings/en/#targetText=Roughly%20one%20third%20of%20the,310%20billion%20in%20developing%20countries)

. In developing nations, food losses and waste amount to $310 billion per year. And fruits and vegetables have the highest wastage rates of any food.

One of the chief reasons for the waste is that it takes just one spoiled product to force disposal of the entire lot. If a needle is discovered in a crate of strawberries, for example, the whole delivery must be thrown away. Meanwhile, fruit branded Grade-A can turn out to be Grade-C or worse, with little recourse for the buyer once the product has traveled halfway around the world, from Indonesia, say, to Wisconsin.

“No traceability means that if you get a bad apple there's no way for you, as a consumer, to complain to the brand and say ‘you guys sent me a bad apple,’” explains Gary Loh, CEO of DiMuto (https://dimuto.io/).

“A digital tracking system was the ‘Holy Grail’ that we needed to find. We wanted to create a solution to register and follow every piece of fruit coming our way, adding up to millions. And we had to put all this data into the cloud. That's where Google came in."

*—Gary Loh, CEO, DiMuto*

Gary's mission in founding the agritech startup last year, after a career in private equity and food distribution, was to leverage blockchain and AI to foster a world where trade is conducted in a fair, sustainable manner. He felt a critical need to cultivate trust between farmers, consumers, and communities. The vision? To tag every orange, avocado, and tropical durian fruit using AI image analysis, and create a unique blockchain record of each product in the cloud.

The aim was to transform rural livelihoods and promote consumer safety and interests from Thailand to Mexico. Gary's team would need the most powerful AI and cloud solutions available. To turn this ambitious idea into a reality, the team turned to Google Cloud (https://cloud.google.com/).

“[Lack of] traceability is a disaster for the farmer. A digital tracking system was the ‘Holy Grail’ that we needed to find,” says Gary. “We wanted to create a solution to register and follow every piece of fruit coming our way, adding up to millions. And we had to put all this data into the cloud. That’s where Google came in.”


## Blockchain solutions for a fair global playing field

When people talk about blockchain, they mostly think of Bitcoin and other cryptocurrencies. Gary realized that blockchain is essentially a powerful journal that can be applied to track avocados as easily as digital coins, where custody of proof is written in a digital ledger that contains an impartial record of each item.

In his earlier years in the food industry, Gary relied on radio-frequency identification (RFID) tagging to ensure shipment integrity. But RFID is mainly an industrial tool related to warehouse management, useless to any buyer who doesn't own an RFID reader.

Gary believed the combination of AI image analysis and blockchain records of each piece of fruit could democratize accountable food sourcing and delivery, making it accessible to smallholder farms in remote corners of the world.

DiMuto developed a system called DAC, or digital asset creation, to stamp every piece of fruit with a QR tag which would enable stakeholders to look up any item in the blockchain record. To make it work, the firm needed the most accurate AI image analysis technology available. It also needed powerful data analytics to ensure a clear view of each link in the supply chain. And it required limitless cloud capacity to store blockchain records of millions of pieces of fruit.

The answer? A combination of BigQuery (https://cloud.google.com/bigquery/), AI Platform (https://cloud.google.com/ai-platform/), AI Hub (https://cloud.google.com/ai-hub/), and TensorFlow (https://www.tensorflow.org/), which provided the machine learning and cloud capacity needed to make DiMuto's vision of fair and transparent agriculture to come true.

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>
> *—Adrian Teo, COO, DiMuto*

“To meet the needs of packing houses all over the world, you need to be able to move from 3G to 4G to ‘no-G’ situations,” says Gary. “Having a partner that knows how to manage cloud situations was critical. Google has huge amounts of experience in this.”

TensorFlow was the critical tool that enabled DiMuto to analyze product images piped through the company’s DAC system, classifying each fruit and assigning a credit score from flawless on one end to defective on the other.

“Essentially, once we have the data on the supply chain, we can use TensorFlow’s immense power to push those insights out," says DiMuto COO Adrian Teo.

The results have been satisfying. Within a year of launching, DiMuto has identified, classified, and tracked 100 million fruits, or roughly $100 million worth of produce, using Google-enabled AI and blockchain technology. DiMuto has built a global network spanning the U.S., Mexico, Australia, Singapore, Indonesia, Malaysia, and Thailand. It plans to expand soon into South America, South Africa, Europe, and the rest of Asia.

## Making things personal through an engaged partnership

A trip to a remote Thai fruit farm brought home to Gary's team how vital Google Cloud would become as a partner in building solutions to challenges such as empowering farmers and overcoming food scarcity.

DiMuto began its digital journey with a competing cloud provider. It made the switch to Google Cloud after meeting the team at the Google Singapore office and being impressed by its “proactive engagement.”

“The team was truly excited about our project, and really understood the issues we wanted to solve,” Gary recalls. “It sees agricultural transparency as a global problem and wanted to dive right in and partner with us.”

Soon after migration, DiMuto learned through a magazine article about Google Cloud science advocate Allen Day, a champion of blockchain solutions. Discovering that Allen was Singapore-based, it reached out to Google Cloud to see if Allen would like to meet up and talk shop.

“The meeting with Allen was set up immediately,” Gary says. “When we met, Allen jumped on our solution; he was very enthused. The next day we were going to visit a farm in Thailand and Allen suggested joining the trip so he could understand exactly what we are doing.”

For Gary’s team, that was proof of the Google commitment to partnering on world-changing solutions: “With our first cloud provider, we felt we were just another account,” says Gary. “Proactive engagement is where Google Cloud stands out from other cloud platforms.”

## Democratizing agritech to help quality farmers thrive

Nestled in California’s fertile San Joaquin Valley, Fancher Creek Packing is a family-run citrus packer that provides quality produce to discriminating buyers. Lacking the scale of California’s agriculture giants, the company run by Jeff and Christiane Pilegard could not afford to tap global enterprise software giants to manage overseas citrus shipments.

Fancher Creek tapped DiMuto for its global tracking needs, which became the Singapore firm’s first foray into the U.S. Today DiMuto’s blockchain solution is enabling the Pilegards to tag their oranges, lemons, and grapefruit for export to Asia, not only to Japan and South Korea, but also developing markets such as Malaysia, Indonesia, and Thailand.

“Fancher Creek is not the most high tech, and it’s not able to hire big enterprise software firms,” says Gary. “But we didn't need to spend lots of money to partner with Jeff and Christiane and move mountains. One reason is the affordable cloud and AI solutions offered by Google Cloud that enable our platform.”

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## Partnering with Google Cloud to explore new and innovative ideas

As Dimuto moves into new markets such as South Africa and Brazil, it also envisions potential applications for its solution in humanitarian food and medicine aid, where deliveries are often compromised by lack of transparency.

For the Dimuto team, one of the factors enabling its quest for positive impact in global agriculture is a shared belief with Google Cloud in the power of creative collaboration. Underpinning the relationship is joint willingness to experiment with ideas outside the comfort zone.

"Now that we know Google Cloud is eager to engage with our innovative 'madman' ideas in agriculture, we're looking forward to working together for the greater good," says Gary with a smile.

## Learn More

Reach out to our team to see how Google Cloud can help your business. (https://cloud.google.com)